

Task 1:

First of all, when pandas parsed the zip file, it read the CARRIER, “NA”, standing for North American Airlines, as a null value. I filled in the respective cells with “NA” by looking at the corresponding CARRIER_NAME, restoring the “missing” information. Then, for MANUFACTURE_YEAR, NUMBER_OF_SEATS, and CAPACITY_IN_POUNDS, I figured that missing values can be replaced by the median value of planes of the same model. This makes sense because airplanes of the same model typically are made around the same time, have a similar number of seats, and are similar in size. This means that this missing data is Missing at Random because it can be explained by the other variables in a way that is systematic. Lastly, I imputed data for AIRLINE_ID by using the max value of other AIRLINE_IDs with the same carrier name, since, through inspection, the AIRLINE_ID is constant across carrier names.

Task 2:

For all of those columns, I decided to standardize by making everything a string dtype, uppercase, and without extra whitespace. This made it easier to use regex to standardize the manufacturers by just the first word. While this function could consolidate unique values in certain scenarios, the “big wins” and the context of this specific dataset override that.

Task 3:

43428 rows of data were removed, which left me with 88885 rows.

Task 4:

The original histograms, more so for the capacity versus the number of seats, are right-skewed (have a long, flat tail on the right). This makes sense because there are probably more commercial airplanes in the sky that aren’t as heavy as huge, industrial airplanes. The number of seats also probably varies like this because there are many industrial airplanes that have exactly 0 seats, whereas commercial airplanes might have a wider spread of a large number of seats.

After the transformation, the capacity histogram looks much more like a normal distribution. However, the number of seats plot almost looks like the opposite—large tails and nearly nothing in the middle.

Task 5:

It seems that the size of aircraft that has the largest non-operational proportion is the medium aircraft. This makes sense to me because there are probably many medium aircraft that have niche purposes. Therefore, with more aircraft of that size, it is more likely that some may be non-operational. Alternatively, more niche airplane sizes, like small and extra large, will need to be called on and used more often than the medium counterparts, playing into this distribution.

I can’t find much online about what the aircraft status actually stands for, so I’m not sure about the analysis of the plot. However, it seems that O dominates for most aircraft sizes except

for medium aircraft, where B dominates. There seems to be some correlation between operating status and aircraft status, but without a clear definition of what aircraft status is, I can't deduce what the correlation is.

Task 6:

Here are the results from my model training:

	Target	Model	Train RMSE	Test RMSE
0	Seats	Linear Regression	11.402099	11.119664
1	Seats	Random Forest	7.357079	8.065016
2	Capacity	Linear Regression	32252.908646	32867.027087
3	Capacity	Random Forest	21410.380012	22731.667911

The RMSE for both Capacity models is extremely large. I'm not sure why. I suspect it could be because I had to drop many columns so that the model would be able to run in a reasonable time, including SERIAL_NUMBER, AIRLINE_ID, and MODEL, some of which may affect the capacity. In general, the order of magnitude for the error is much higher than I expected.

For each target, the random forest performs better than the linear regression. This could be because the data is nonlinear, and random forest is better at capturing nonlinear relationships than linear regression.

The train RMSE is consistently lower than the test RMSE, except for the linear regression model for seats. However, the difference is very low (<0.2).

Generative AI Statement:

The use of Gemini was helpful to solve some of the issues I was encountering. Some use cases included ReGeX expressions and plotting syntax. Some functions like quantile and crosstab were extremely helpful to learn, and I will be applying them in future work. I also had it diagnose the problem of my model taking really long to train (>45 minutes). By dropping more columns and changing the random forest initialization, it was able to drop training to just a few minutes.